

Probabilistic graphical models

1 Introduction

2 The Noisy-OR Model

2.1 Motivation

In a Bayesian network, a node Y with k binary parents $X_1, \dots, X_k$ needs a conditional probability table (CPT) specifying $P(Y \mid X_1, \dots, X_k)$ for all 2^k parent configurations. This grows exponentially in the number of parents, which is both hard to elicit from experts and expensive to store.

The noisy-OR model is a canonical example of a *compact* (parametric) CPT: it assumes each parent X_i independently has some chance of *causing* $Y = 1$, and Y is on if *any* of the causes succeed. This reduces the number of parameters from $O(2^k)$ to $O(k)$.

2.2 Setup

Each parent X_i is a binary cause of the binary effect Y . Introduce independent “inhibitor” variables: even when $X_i = 1$, the cause is inhibited (fails to trigger Y) with probability q_i . Equivalently, X_i succeeds in causing Y with probability $1 - q_i$, independently of the other causes.

$Y = 0$ iff every active parent’s cause is inhibited. So

$$P(Y = 0 \mid X_1, \dots, X_k) = \prod_{i: X_i=1} q_i, \quad P(Y = 1 \mid X_1, \dots, X_k) = 1 - \prod_{i: X_i=1} q_i.$$

Only k parameters $q_1, \dots, q_k$ are needed, one per parent.

2.3 Leak probability

In practice Y can sometimes be true even when all listed parents are 0 (an unmodeled or background cause). This is handled with a *leak* parameter $q_0 \in (0, 1]$, treated as an always-active virtual parent:

$$P(Y = 0 \mid X_1, \dots, X_k) = q_0 \prod_{i: X_i=1} q_i.$$

2.4 Example

Suppose Y = “fever” has parents X_1 = flu, X_2 = malaria, X_3 = cold, with inhibition probabilities $q_1 = 0.4$, $q_2 = 0.1$, $q_3 = 0.7$ (so flu causes fever with probability 0.6, malaria with 0.9, cold with 0.3). If a patient has flu and a cold but not malaria ($X_1 = 1, X_2 = 0, X_3 = 1$):

$$P(Y = 0 \mid X_1 = 1, X_2 = 0, X_3 = 1) = q_1 q_3 = 0.4 \times 0.7 = 0.28, \quad P(Y = 1 \mid \cdot) = 0.72.$$

2.5 Properties

- **Monotonic:** adding an active cause can only increase $P(Y = 1)$, since it multiplies another factor $q_i \in (0, 1]$ into $P(Y = 0)$.
- **Symmetric in the causes:** the causes contribute independently; order doesn't matter, only which subset is active.
- **Special case:** if every $q_i \rightarrow 0$, this reduces to the deterministic logical OR of the parents.
- **Generalizations:** noisy-AND (dual model, symmetric using De Morgan's law) and noisy-MAX (extension to multi-valued, non-binary variables).